August 1, 23

Nucleoside ⟶ Base + Sugar

Anomeric Carbon
(α, β)

When sugar in not attached to Base: Hemiacetal
Reducing sugar

Nucleoside: Non-Reducing sugar

Reducing sugar:
Hemiacetal থেকে ভাঙলে সহজে Aldehyde এ back করা যায়।

Forms ⟶ Acetal থেকে যায় না।
Non sugar
Reducing

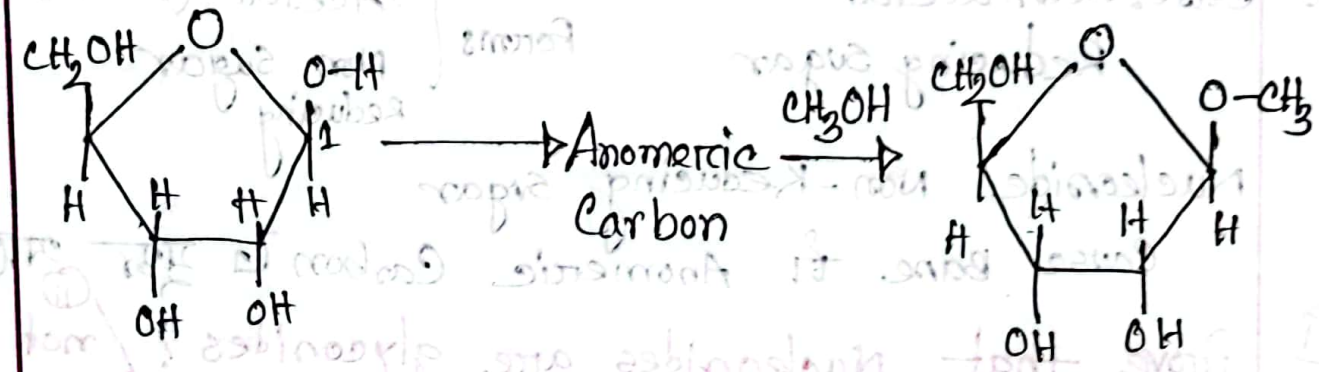
Cause Base টা Anomeric Carbon এ যুক্ত আছে।

Ques. (I) Prove that Nucleosides are glycosides? / (II) not reducing?

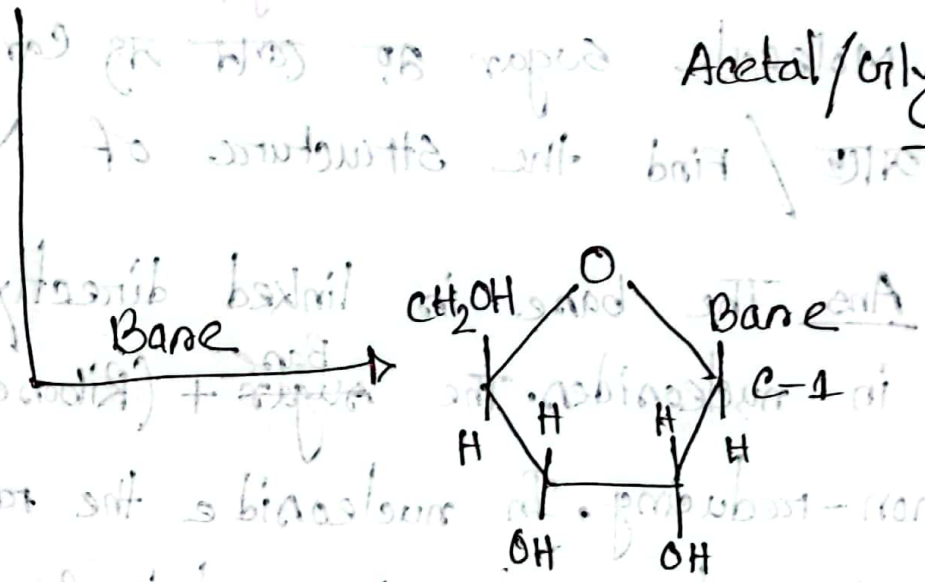
(II) Base Molecule sugar এর কোন নং Carbon এ যুক্ত আছে / Find the structure of Nucleoside?

Ans. The base is linked directly to the sugars in nucleosides. The ~~sugar~~ Base + (Ribose / Deoxyribose) are non-reducing, In nucleoside the ribose sugars is not in the hemiacetal form. The sugar part of the nucleoside remains in glycosidic

form. That in the -OH of the anomeric carbon is not free it forms bond with the base and thus form nucleoside.




Acetal/Glycoside form

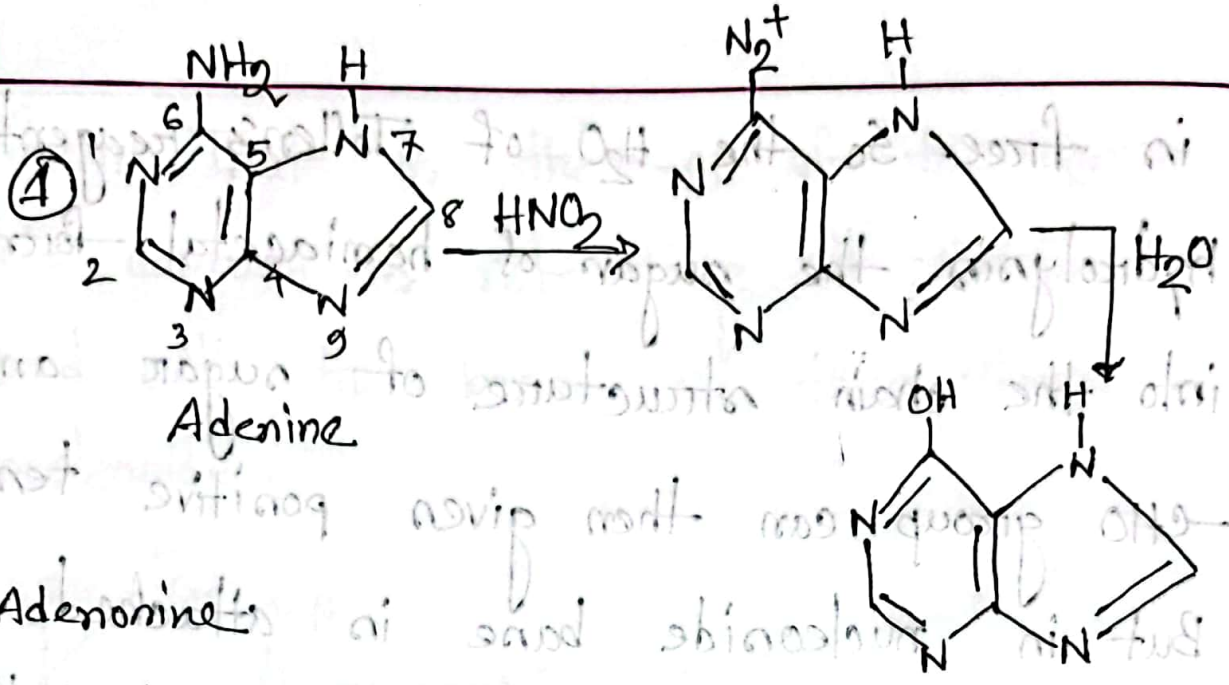



The sugar normally remains in the hemiacetal form. where the C-1, anomeric carbon (OH)

in free. So the $H_2O$ of Tollen's reagent can hydrolysis the sugar of hemiacetal form. into the chain structure of sugar and the -CHO group can then given positive test.

But in nucleoside base is attached with sugar. It is the C-1 of sugar with which the base is attached. The base removes the 'H' of C-1 OH and makes the acetal, non-reducing sugar.

# In which position of Base the sugar is attached?



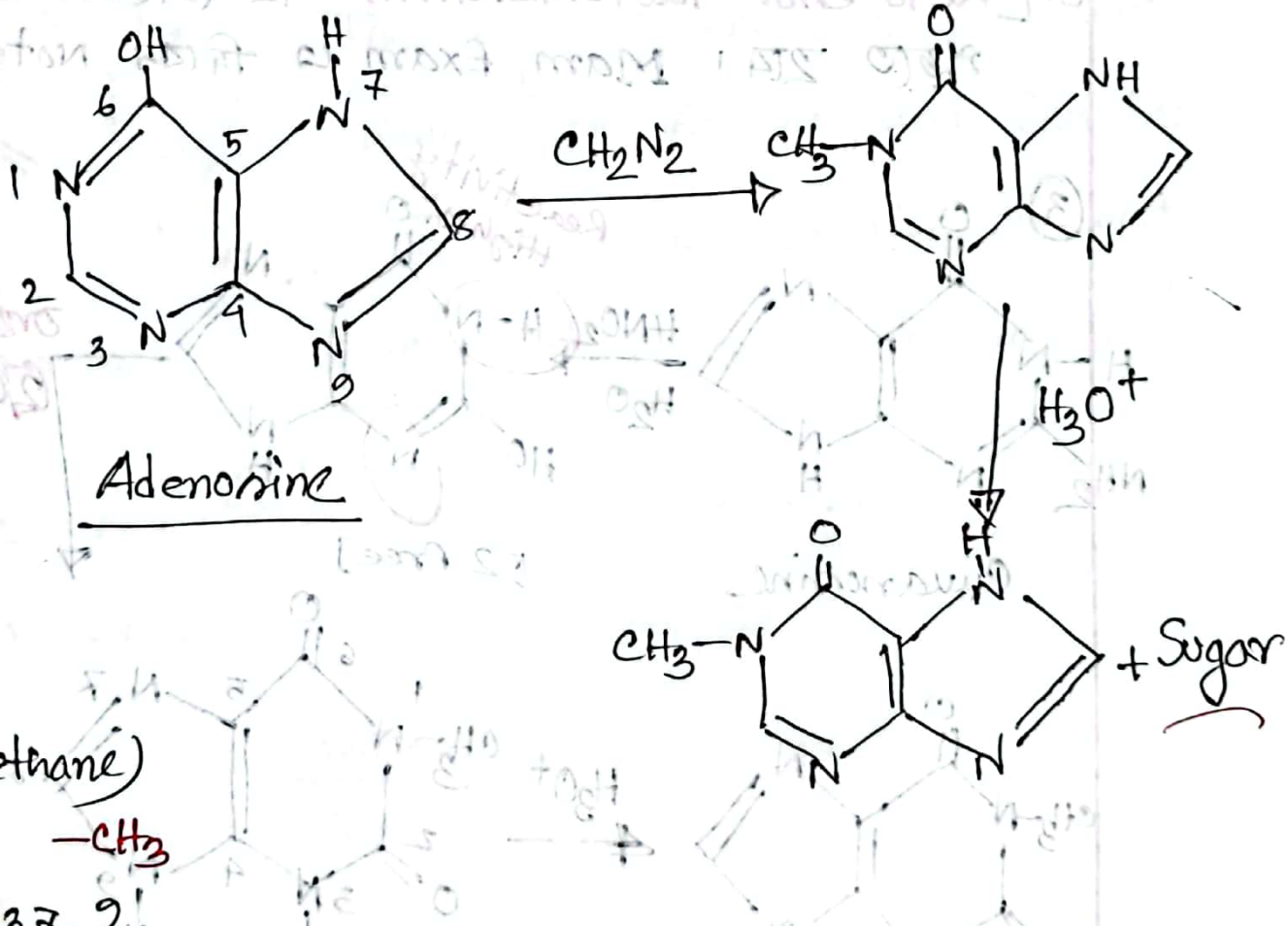

Adenonine

↳ Adenine + Ribose

When Adenine is treated with $HNO_2$ the $NH_2$ group in 6-C is deamoniated and evolve $N_2$ gas at room temperature in presence of water and replace $NH_2$ at C-6 by OH group. So the 1 $NH_2$ is not attached with the sugar.

② When the phenol derivative is

treated with diazomethan ($CH_2N_2$) the $CH_3$ group attack at C-1. Though the methyl group can attached at C-1 of the base so the C-1 was free and there is no sugar attached.




$CH_2N_2$
(Diazomethane)



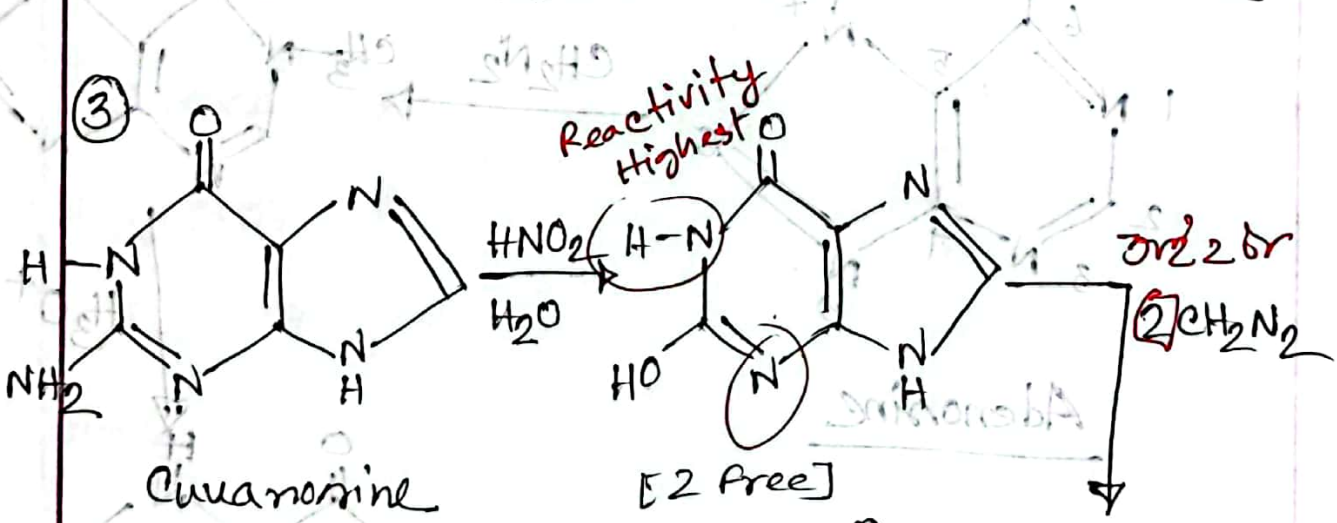

Why C-1?

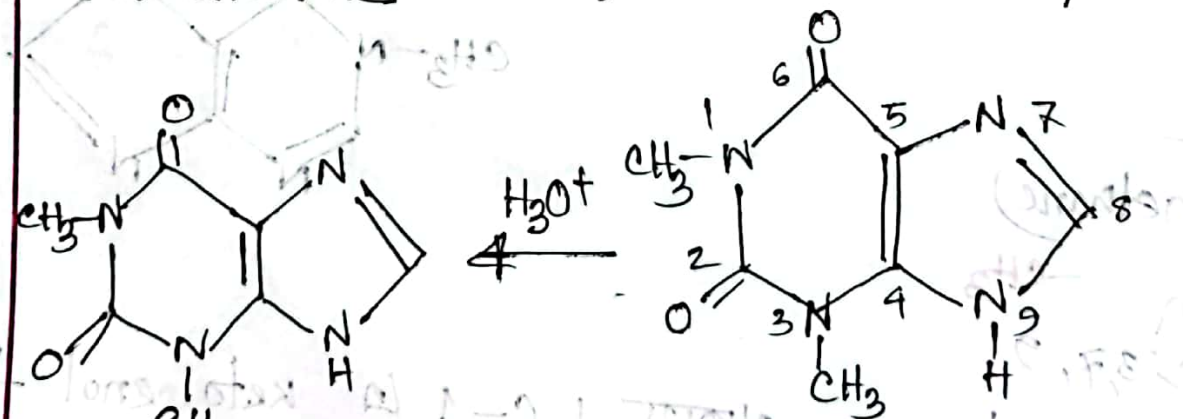
phenolic -OH থাকায় C-1 এ keto enol tautome rism হয় এজন্য 3,7,9 C এ সুগার না হয়ে C-1

এ যুক্ত হয়। 3, 7, 9 এ N এর reactivity এর 1 এর তুলনায়,

After hydrolysis, this N-$CH_3$ group remains unchanged, as the sugar is not attached to it.

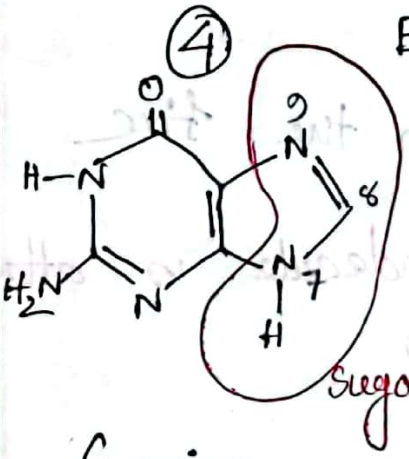
**[Keto enol Tautomerism বই থেকে Note করে পড়তে হবে। Mam Exam এ দিবেন Noted] **



Though the two methyl group is attached at C-1 and C-3 position. So the C-1, C-3 and C-2 were free.

Aug 13, 2023




Guanine

Enzyme reacts in a specific site. The sugar attached in the same position of Adenine and Guanine. In a same position of both base the sugar in attached, so the equivalent and non-equivalent position in two bases are given below:

| Guanosine | | Adenosine |
|---|---|---|
| N-1 | ≠ | N-1 |
| C-2 | ≠ | C-2 |
| N-3 | ≠ | N-3 |
| C-6 | ≠ | C-6 |

The sugar is not attached with the six membered ring of the base.

So the Equivalent positions in two bases are

N-7 ≈ N-7

N-9 ≈ N-9

C-8 ≈ C-8

The sugar is attached in the five membered ring.

Interprete that sugar molecule is attached to the 5-membered ring?

Ans: 1 to 4 point.

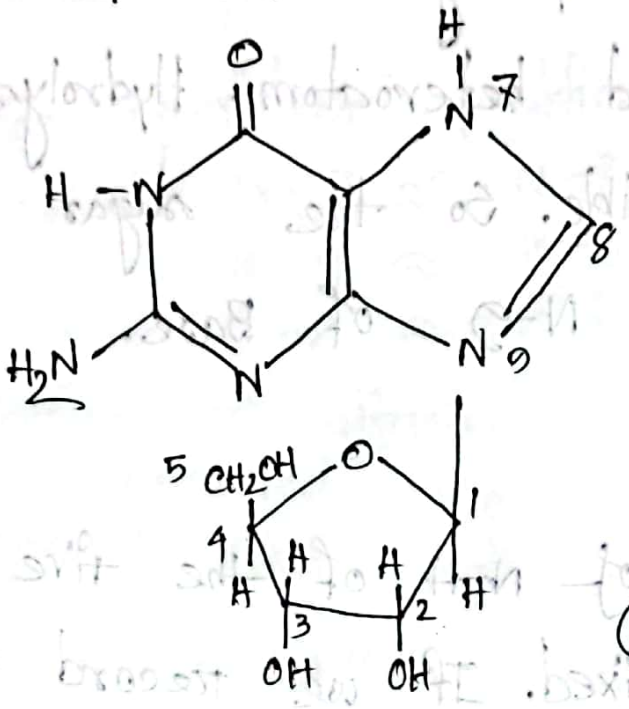
When the nucleoside is hydrolysed with water ($H_2O$) the hydrolysis must occur between the carbon and heteroatom. Hydrolysis of C–C bond is impossible. So the sugar is attached with N-7 or N-9 of Base.



The H of N-H of the five membered ring is not fixed. If we record UV the $\lambda_{max}$ for nucleoside Guanosine is 463 nm and if the N-9 is replaced by $CH_3$ the (Methylation) $\lambda_{max}$ is 460 nm.

But if the N-7 is replaced by $CH_3$ the $\lambda_{max}$ = 440 nm. ( 460 ≈ 463. So N-9 position in more equivalent than N-7)

So the sugar is attached at N-9 position of Base.




Guanine + Ribose = Guanosine

Fig. Final structure of Nucleoside when the base is purin.

Primary structure Exam

Exam

## Position of Phosphate Group.

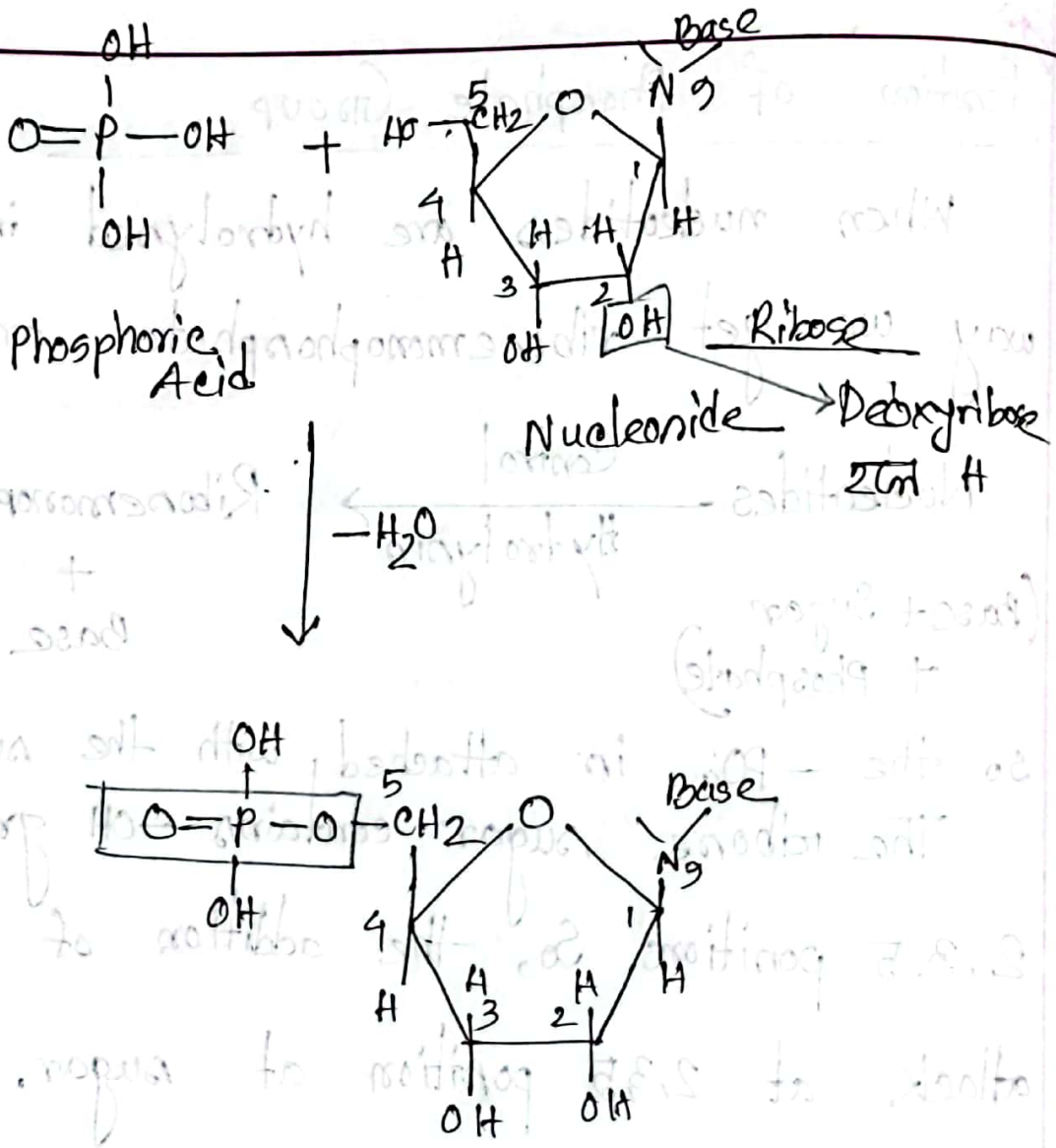
When nucleotides are hydrolysed in a controlled way we get ribonemonophosphate and Base.

Nucleotides $\xrightarrow[\text{Hydrolysis}]{\text{Control}}$ Ribonemonophosphate + Base

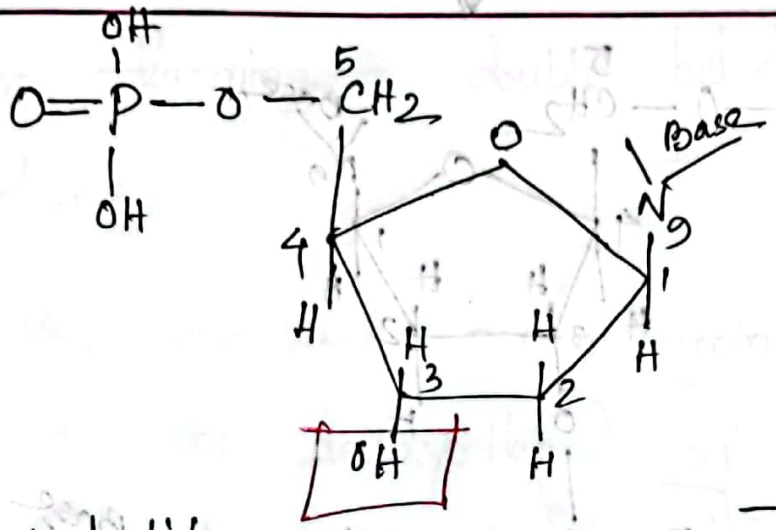
(Base + Sugar + Phosphate)

So the $-PO_4$ is attached with the sugar part.

The ribose sugar contains $-OH$ group at 2, 3, 5 positions. So, the addition of $H_3PO_4$ will attack at 2, 3, 5 position at sugar.

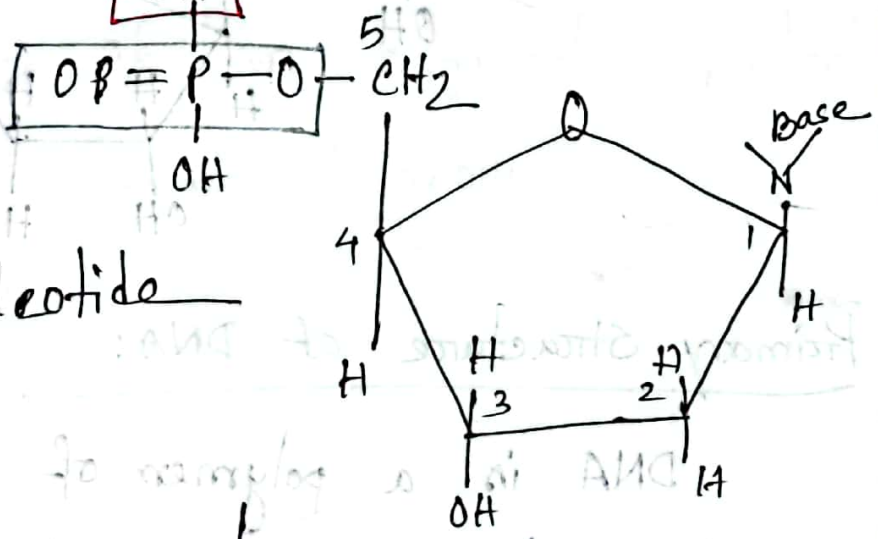



Base প্রথমবার্ draw করতে হবে। পরে Base লিখলেই হবে।

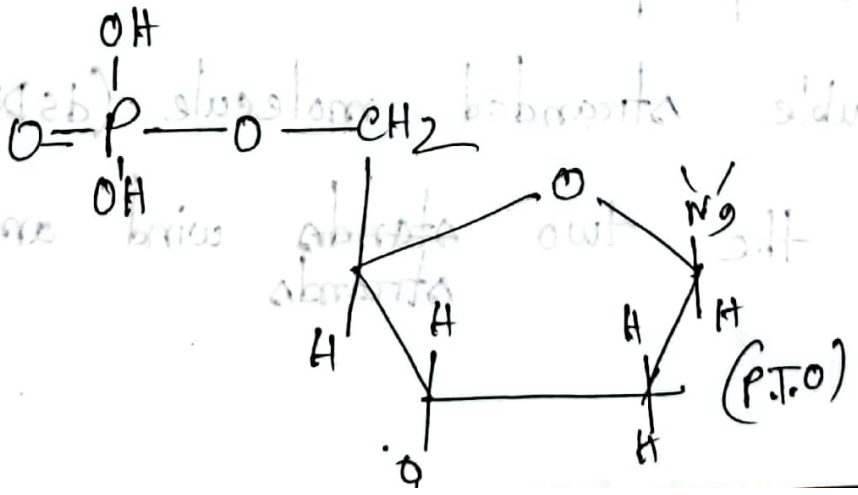
① 


Nucleotide

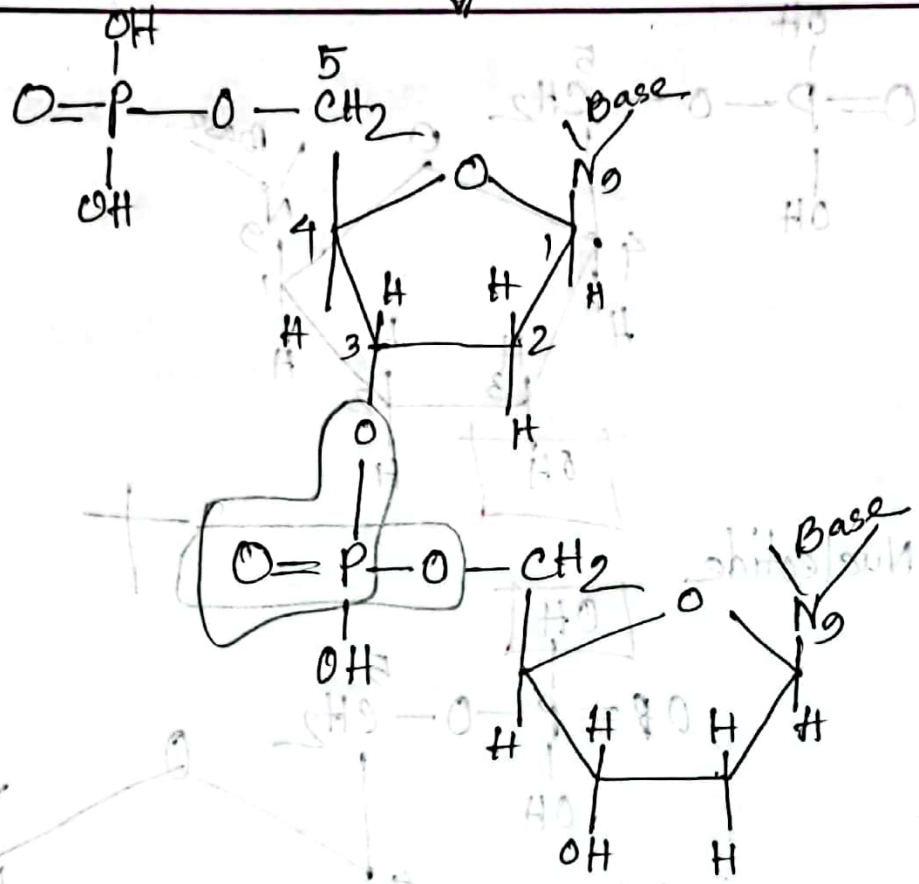
+

② 


Nucleotide




(P.T.O)






.DNA

Aug 14, 2023

## Primary Structure of DNA:

DNA is a polymer of deoxyribonucleoside monophosphate (dNMP) covalently linked by 3′ ⟶ 5′ phosphodiester bonds. DNA exists as a double stranded molecule (dsDNA) in which the two ~~stands~~ strands wind around each

other forming a double helix (secondary structure).

Phosphodiester bonds join the 3' hydroxyl group of the deoxyribose of one nucleotide to the 5' hydroxyl group of the deoxypentose of an adjacent nucleotide through a phosphoryl group.

## Secondary Structure / Whatson and Crick Model :

## Secondary structure of DNA:

In the double helix, the two chains are coiled around a common axis called the helical axis. The chains are paired in an antiparallel manner (i.e.) the 5' end of the one nucleotide strand is paired with the 3' end of the other nucleotide strand.

In the DNA helix, the hydrophilic deoxyribose phosphate backbone of each chain is on the outside of the molecule, where as the hydrophobic bases are stacked inside the overall structure resembles a twisted ladder. The bases of one strand of DNA

are paired with the bases of the ~~second~~ bases of the second strand by hydrogen bonding.

- Helical axis: Base গুলো H-Axis এর সাথে এক পাশ দিয়ে 90° Angle করে অবস্থান করে।
- 3D structure
- Deoxyribose phosphate (Backbone)

Why?

A=T তাই সহজ। এখান থেকেই জটিল শুরু হয়।

Semi-conservative pathway / Replication:

↓

বলা হয়। কারণ strand এর একটা part আসে Mother DNA থেকে। Body থেকে Base আসে depending on the part of the Mother DNA (A থাকলে T, G থাকলে C and vice versa).



## Replication of DNA

When the two strands of dsDNA are separated, each serve as a template for the replication (synthesis of a new complementary strand, this produces two daughter molecules each of which contains two DNA strand (one new, one old), in an antiparallel orientation. This process is called semiconservative replication because, although the parental duplex is separated into two halves, and therefore is not conserved as an entity, each of the parental strands remains ~~exact~~ intact in one of the two new duplexes.

Origin of replication starts from A=T unwind and formation of replication fork by Helicases enzyme.

HDP (Helix destabilizing protein) don't let the single strand to form each other.

→ ম্যাম বই পড়তে বলেছেন (strainer এর)

Book name: F. ~~Sty~~ Stryer +

Assignment: Finate vol 2 থেকে Pyrimidine Base এর কোন position এ sugar molecule যুক্ত আছে।

Biochemistry Denine-R. Ferrier (Mainly)

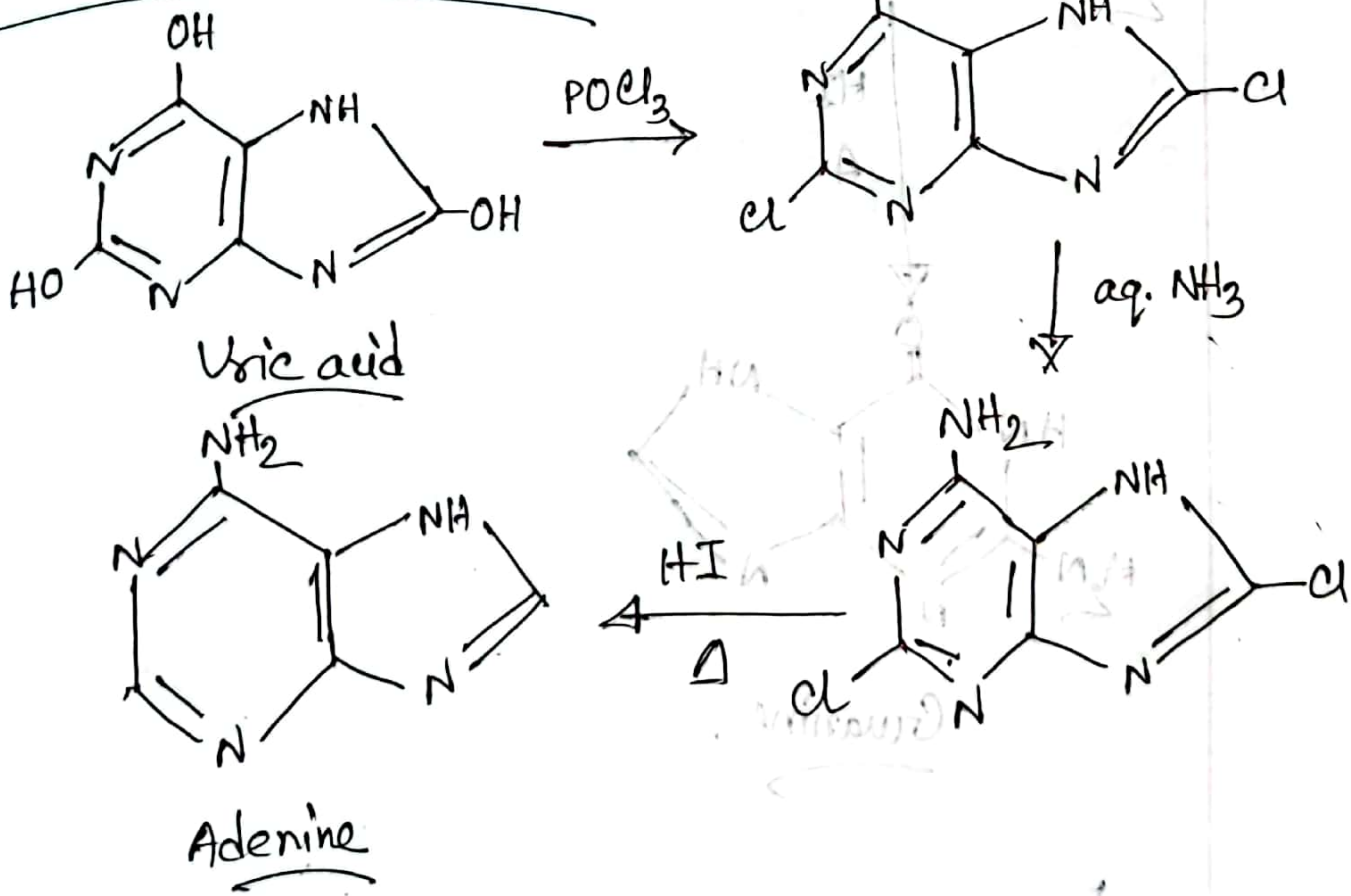
20/8/23

## Chargaff Rule:

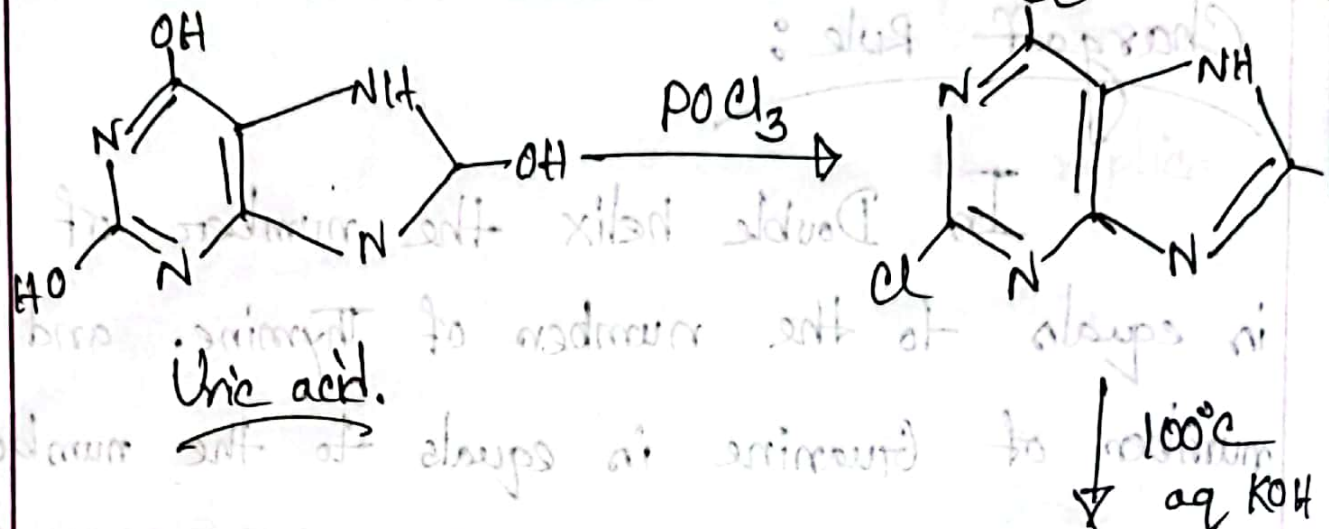
In Double helix the number of Adenine is equals to the number of Thymine. and the number of Guanine is equals to the number of cytosine. So the number of purin base is also equal to the number of pyrimidine base.

**Ques.** Adenine from uric Acid?

### Synthesis of Adenine:

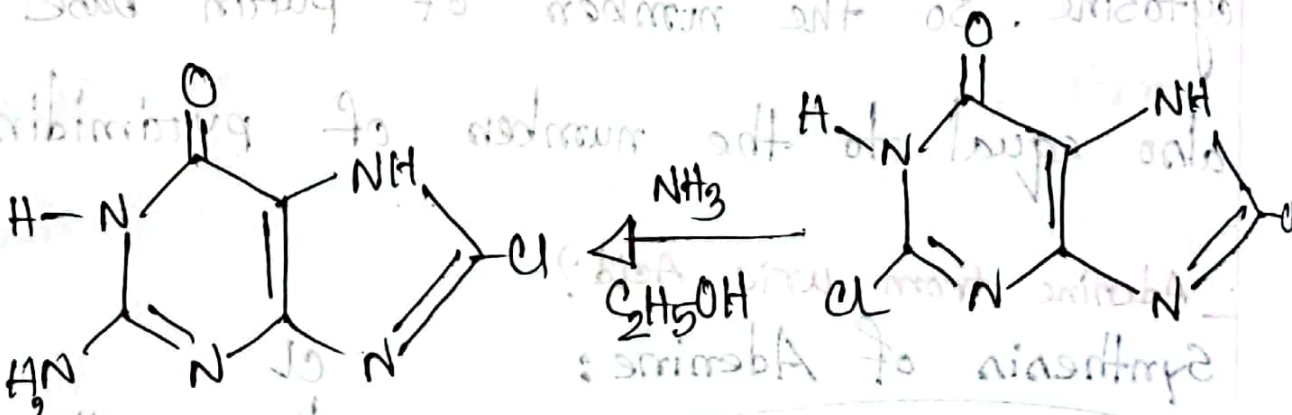


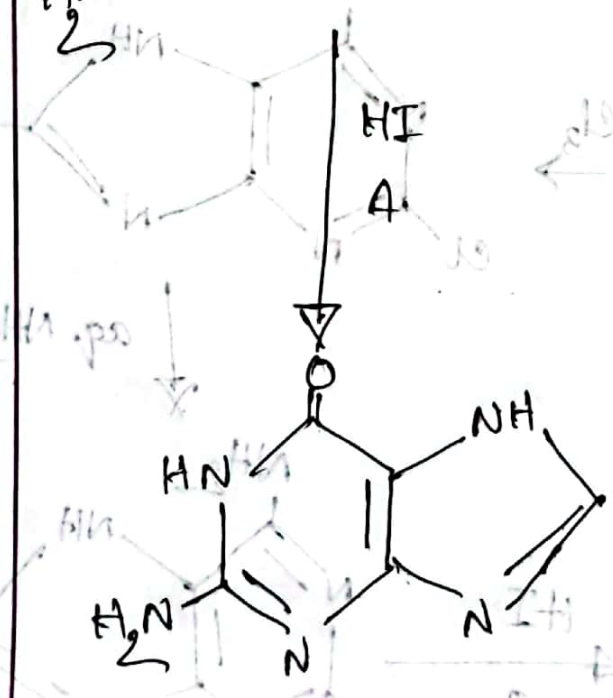
# Synthesis of Guanine:




Uric acid.



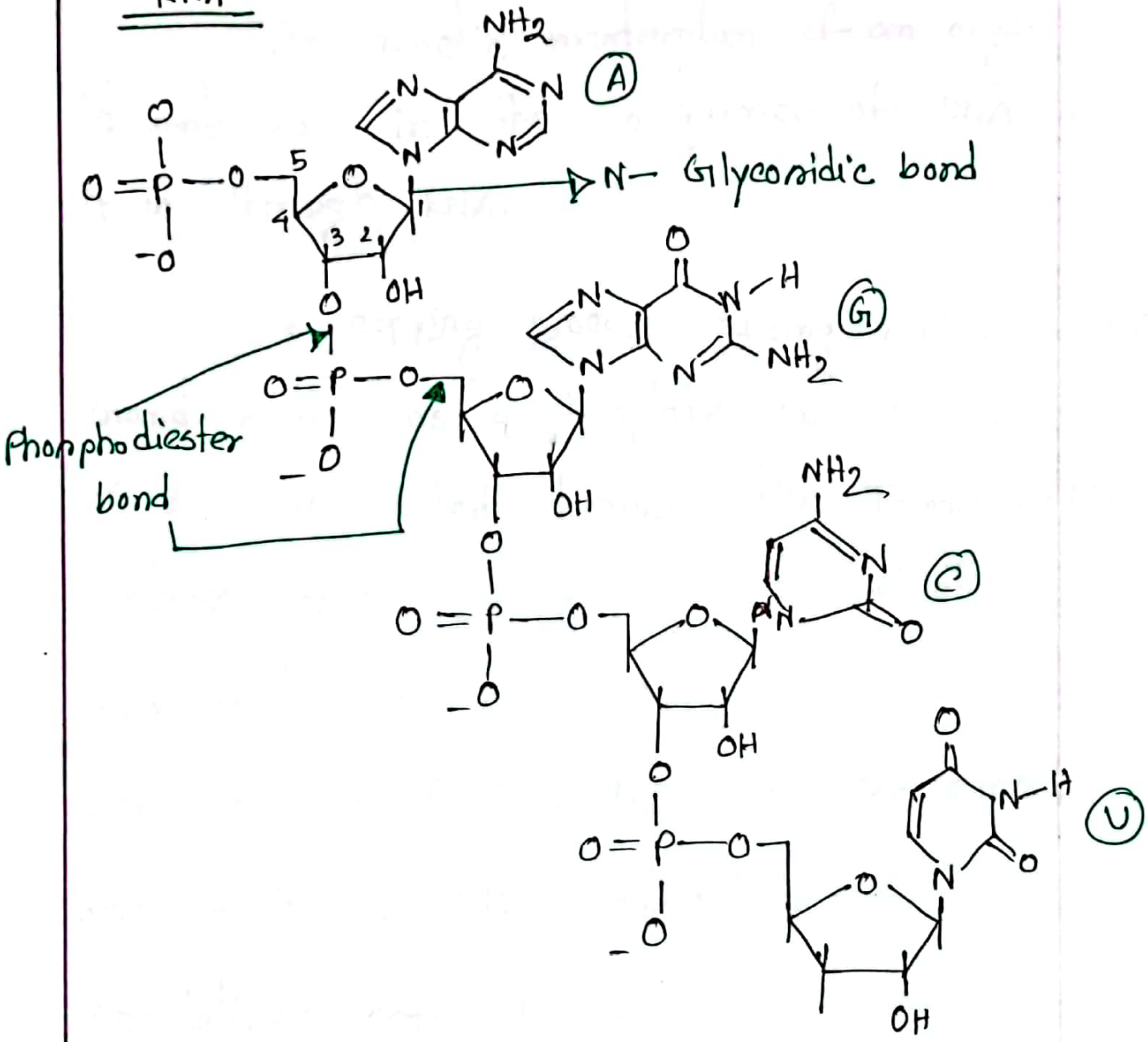



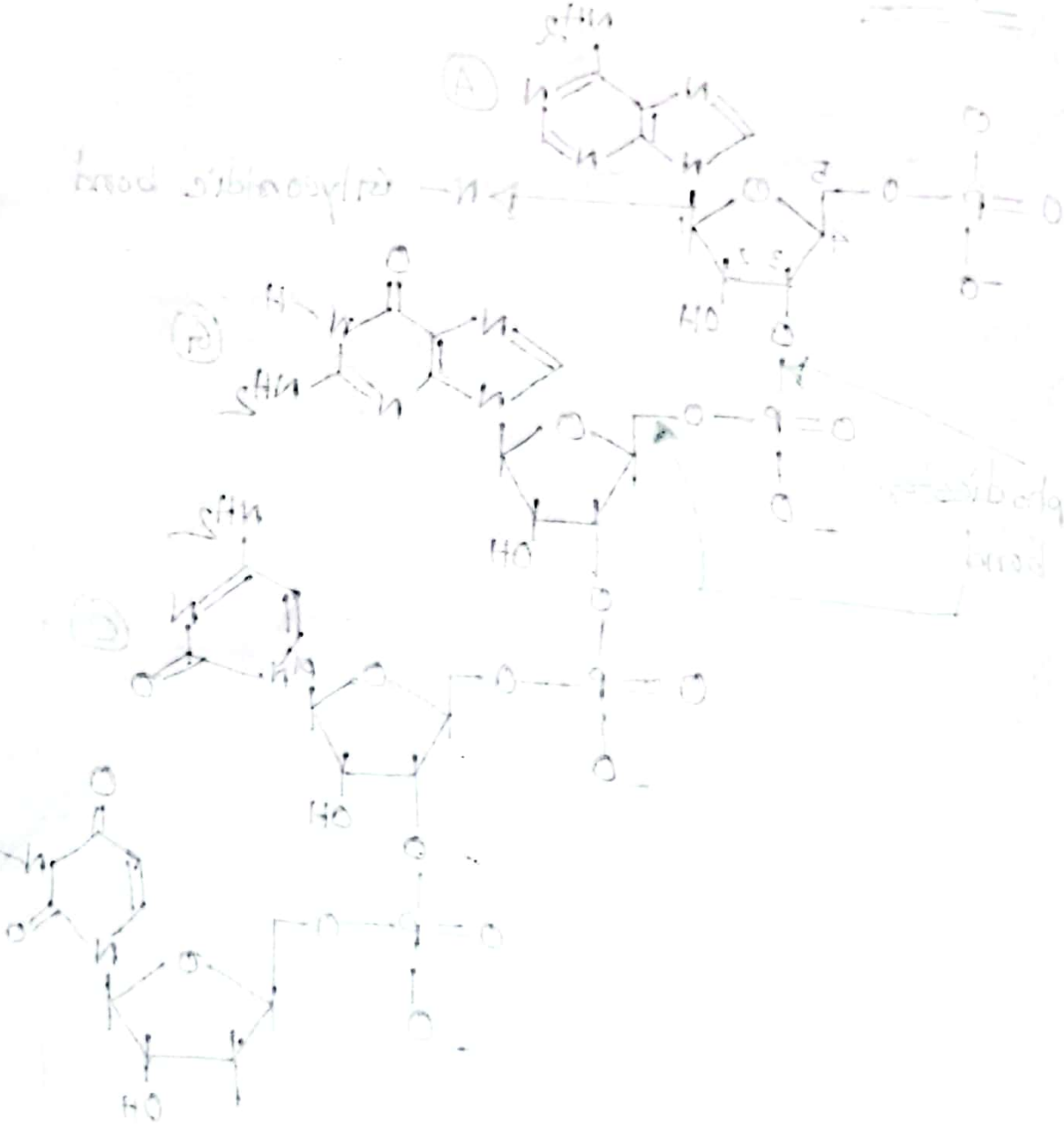




Guanine.

21/8/23

## RNA :





Difference between DNA and RNA.
(পড়তে হবে)

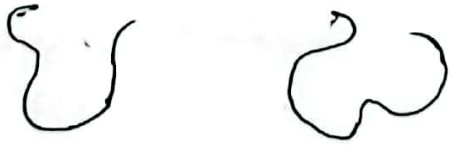
3B= 1C = 1AA

The genetic masterplan of an organism is contained in the sequence of DNA. but it is through RNA.

The copying process during which a DNA stand serves as a template for the synthesis of RNA is called transcription. Transcription produces messenger RNA (mRNA), which are translated into sequences of amino acids and ribosomal RNA (rRNA) transfer RNA (tRNA) and additional RNA molecules that perform specialized catalytic and regulatory function and are not translated are called non-coding RNA (nc. RNA).

**rRNA:** rRNA are found in association with several proteins as components of the ribosome, the complex structure that serves as the sites for "protein synthesis.

**tRNA:** tRNA are the smallest of the three major types of RNA molecule. There are at least one specific type of tRNA molecule for each of the 20 amino acids. It also serves as an adaptor to enter the specific amino acid.



**mRNA:** mRNA occupies only 5% of the total RNA in the cell. mRNA is copying RNA in that it carries the entire genetic information from DNA for use in protein synthesis.

Primary structure for exam
Structure Draw করতে বলতে Base এর structure
আর + Numbering আর Draw করতে হবে Exam এ।

Difference of DNA, RNA must পড়তে হবে।